

This report outlines the theoretical foundation and algorithmic framework for a **Proactive Virtual Keyboard Nudge System**. This system is designed for submission to **ACM-CHI 2026** and focuses on the intersection of Cybersecurity and Human-Computer Interaction (HCI).

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# Project Proposal: Proactive Geometric-Probabilistic Password Nudging

**Target Venue:** ACM-CHI 2026 (Privacy, Security, and Visualization)

**Core Innovation:** A real-time, client-side virtual keyboard that dynamically blocks keystrokes based on high-dimensional vector similarity and conditional probability.

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## 1. Executive Summary

Current password security relies on **reactive** feedback (meters) or **rigid** policies (Complexity Rules). This system introduces a **proactive intervention** that curates the user's input space in real-time. By utilizing a "Dual-Guardrail" approach—combining **TF-IDF Vector Distance** and **Next-Char Conditional Probability**—the system mathematically guarantees that a user cannot create a password that exists within the "vulnerability manifold" of known data breaches.

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## 2. The Problem Landscape

Despite complexity requirements, users follow predictable "mangling" habits (e.g., replacing 's' with '\$').

- **Exact Blacklists** are easily bypassed by minor character changes.
  - **Probabilistic Meters** are reactive; they tell the user a password is weak *after* the effort of creation, leading to frustration and "security fatigue."
  - **Search Space Vulnerability:** Attackers use optimized Markov models to guess passwords based on human habits.
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## 3. Proposed Algorithmic Framework

The system calculates an **Integrity Score (\$\$\$)** for every potential keystroke  $sk$ . If the score falls below a threshold, the key is visually disabled or "nudged."

## A. Parameter 1: Vector Proximity ( $D_v$ )

- **Method:** TF-IDF Weighted Character n-gram Embeddings.
- **Function:** Projects the partial string into a high-dimensional vector space.
- **Intricacy:** TF-IDF (Term Frequency-Inverse Document Frequency) weights the importance of character sequences. It de-emphasizes common patterns (like 123) and amplifies unique ones.
- **The Goal:** To maintain a maximum **Cosine Similarity** radius from "centroids" of millions of leaked passwords (e.g., *RockYou* dataset).

## B. Parameter 2: Next-Char Probability ( $P_{next}$ )

- **Method:** Variable-Order Markov Chains derived from breach datasets.
- **Function:** Calculates the probability  $P(k | \text{history})$  that the next character  $k$  follows the current sequence.
- **Derivation (Shannon Entropy):** Based on the principle of Surprisal.

$$I(k) = -\log_2 \left( \frac{1}{P(k)} \right)$$

- **The Goal:** To prevent "predictable jumps" (like keyboard walks or common structural tropes) even if the resulting string is not in a blacklist.

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## 4. The Integrated "Integrity Score" Formula

The system assigns equal weight to both parameters for the initial implementation:

$$B_s = 0.5 \cdot \text{Sim}(\vec{V}_{current+k}, \mu_{leaks}) + 0.5 \cdot P(k | \text{Blacklist\_Data})$$

- **High  $B_s$ :** The key is blocked because it either moves the password into a "Leaked Habit Cluster" or follows a "High Probability/Low Entropy" path.

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## 5. Formal Proof of Algorithmic Superiority

To validate the system without user experiments, we employ **Simulation-Based Technical Evaluation**:

1. **Search Space Contraction:** By blocking the top  $n$  most likely characters at each step, the algorithm reduces the "vulnerable search space" by several orders of magnitude while maintaining enough "available keys" for user agency.
2. **Structural Divergence:** Using t-SNE visualizations, we can demonstrate that passwords generated under these constraints occupy "Sparse Regions" of the vector space, far from the "Dense Clusters" where attackers focus their brute-force efforts.
3. **Resistance to Optimized Guessing:** Unlike standard passwords, those generated with

this system have a near-flat probability distribution, maximizing **Guessing Entropy (\$G\$)** against Markov-based cracking tools like *HashCat*.

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## 6. Comparison Table

| Feature      | Standard Keyboard   | Reactive Meter (zxcvbn) | Proactive Dual-Metric System |
|--------------|---------------------|-------------------------|------------------------------|
| Intervention | None (Post-hoc)     | After typing (Reactive) | During typing (Proactive)    |
| Logic        | Boolean Rules       | Pattern Matching        | Geometric + Probabilistic    |
| Leak Defense | None                | Exact/Partial Match     | High-Dim Vector Proximity    |
| Search Space | Infinite (but weak) | Restricted by rules     | Optimized for Entropy        |

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## 7. Key Research Literature for Presentation

- **Telepathwords (Microsoft Research):** The first to use probability nudges; our system improves this by adding Vector Distance.
  - **Zxcvbn (Dropbox/Low-Budget Strength):** The industry standard for pattern-based guessing; our system moves this logic to a proactive UI.
  - **Shannon (1948) - A Mathematical Theory of Communication:** The foundation for our Entropy and Information Theory claims.
  - **Habib et al. (2017) - Password Creation in the Presence of Blacklists:** Justification for why "exact matching" fails and similarity-based blocking is required.
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### Next Steps for Development:

1. **Centroid Selection:** Identify the 10 core "Weak Habits" clusters from the *RockYou* dataset.
  2. **Latency Optimization:** Implement Locality Sensitive Hashing (LSH) to ensure the vector comparison happens in  $<100\text{ms}$  on the client side.
  3. **Availability Proof:** Run a Monte Carlo simulation to determine the optimal "Blocking Threshold" that balances security with keyboard usability.
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